MAIN PRODUCTS

Sugar Alcohol & New Sugar Source

- Erythritol
- Allulose
- Blended Sweeteners
- Crystalline Fructose
- Trehalose

Starch & Modified Starch

- Modified starch for mochi bread F110 series
- Modified starch for crystal skin F120 series
- Modified starch for meat products F100 series
- Modified starch for baking sauces B151 series
- Modified starch for dairy products B151 series/ B153 series
- Modified starch for sauce B151 series / B162 series

Organic Products

- Organic Erythritol
- Organic Isomaltooligosaccharide
- Organic Fructooligosaccharide
- Organic Resistant Dextrin
- Organic Corn starch
- Organic Maltodextrin
- Organic Glucose Syrup
- Organic Crystalline Fructose

Corn Byproduct

- Corn Steep Liquor
- Corn Gluten Meal

Tapioca Starch Processed Products

Prebiotics Series

- Isomaltooligosaccharide (IMO)
- Galactooligosaccharide (GOS)
- Fructooligosaccharide (FOS)
- Human Milk Oligosaccharides (HMOS)
- 1-Kestose

Dietary Fiber

- Polydextrose
- Sugar-free Polydextrose
- Resistant Dextrin

Starch Sugar

- Maltotetraose
- Maltodextrin
- Corn Syrup Solids
- Dextrose Monohydrate
- Dextrose Anhydrous
- Liquid Glucose
- HFCS-42/55
- High Maltose Syrup/Powder

Recommended products

- DHA (docosahexaenoic acid)
- Isomerized lactose (4-O- β -D-galactose-D-fructofuran)
- Human milk oligosaccharides（HMOS）

Stock listed code：002286



Create better life by biotechnology

Seek profit for cultivators Bring health to life

BAOLINGBAO BIOLOGY CO.,LTD.

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保龄宝生物
BAOLINGBAO BIOLOGY

COMPANY PROFILE

Baolingbao Biology Co.,Ltd., a leading manufacturer of corn-based health and food ingredients, is located in the center of North China Plain near the historical city Jinan in Shandong Province of China. Baolingbao has been using innovative technology to transform maize and sugar into value-added ingredients for customers in the food, dairy, beverage, pharmaceutical, animal feed and cosmetics.

Baolingbao was established in 1997 and listed in Chinese stock market in 2009. The initial product is IMO (Isomaltooligosaccharide) which is a good prebiotic. Over the years, many new type products had been developed such like Erythritol, Resistant Dextrin, Crystalline Fructose, Allulose, Polydextrose, FOS, GOS, etc. Today, Baolingbao produces a wide range of products for fields of foods and beverages, pharmaceuticals, cosmetics, etc. Always striving for higher quality, Baolingbao's products are well accepted by customers.

"Seek profit for cultivators Bring health to life" is Baolingbao's mission. Baolingbao is trying the best to learn cutting-edge technology, and then it adds the value to the products by the technology. That meets the wishes of cultivators. At the same time, Baolingbao provides the health products to the society.

Become the **L**eader in **B**iotechnology

Built to **L**ast in the **B**usiness



WE ALWAYS FOCUS ON EVERY PROCESS



ISO9001:2000 · IP · ISO22000 · HALAL · U · USDA ORGANIC · SGS · ISO14001 · NON GMO Project VERIFIED · SMETA

DIETARY FIBER

■ Polydextrose

Polydextrose is a polysaccharide composed of randomly cross-linked glucose units with all types of glycosidic bonds. It is an excellent water-soluble dietary fiber.

- Produces short chain fatty acids to stimulate the bowel function
- Prebiotic properties
- Solubility is about 80%
- Appropriate viscosity
- Freezing point depression
- Low calorie value 1Kcal/g
- Well tolerated
- Low glyceamic response

Sugar-free Polydextrose

- Better taste
- Clear and transparent syrup, easy to match with other colors
- Lower viscosity than conventional polydextrose
- Non-reducing, sugar-free, low sweetness

■ Resistant Dextrin（Soluble Fiber）

Resistant dextrin is a light yellow syrup or powder product. It is a soluble fiber of natural origin, derived from starches. It is easily soluble in water for a transparent solution with lightly sweet, convenient processing. It can be used in many kinds of foods, beverages and nutritional products. But it has high stability and does not affect their original flavor.

- Improve lipid metabolism
- Promote mineral absorption
- Improve bowel movements to prevent constipation
- Inhibition of postprandial blood glucose rise



PREBIOTICS

■ Fructooligosaccahride (FOS)

FOS, derived from sugar, is another kind of prebiotics. FOS is mainly composed of the three sugars: kestose (GF2), nystose (GF3) and fructofuranosylnystose (GF4)

- Excellent prebiotics properties
- Helps intestinal minerals and vitamins absorption

■ Isomaltooligosaccharide (IMO)

IMO, derived from refined corn starch, is a kind of prebiotics. IMO is a mixture of saccharides including isomaltose, panose, isomaltotroise etc.

- Regluates & Improves intestinal flora
- Helps intestinal minerals absorption
- Stable in the environment of heating and/or acid
- Keeps moisture
- Maillard reaction

■ Galactooligosaccharide (GOS)

GOS, derived from lactose (milk sugar), is a kind of natural functional oligosaccharides. With their indigestible nature, GOS is also classified into prebiotics. It has been successfully used in a wide variety of formulations such as infant nutrition, grown up milk, dairy products and beverages. It's marketed in a syrup form but could also be applied in combination with other components in a co-spray dried form.

- High solubility
- Heat and acid stability
- Low Glycemic index
- Low calorie
- Non-digestible
- Promoting growth of bifidobacteria

■ Isomerized lactose(4-O-β-D-galactose-D-fructofuran)

A disaccharide consisting of galactose and fructose linked by a β-(1-4) glycosidic bond, also known as lactulose. It is produced by isomerizing lactose, which is naturally present in milk.
Industrial production of isomerized lactose is produced by heat treatment of lactose as raw material under alkaline conditions.
Regulate intestinal flora - isomerized lactose intervention increases the number of probiotics, optimizes the intestinal microenvironment, and relieves constipation
Reduction of intestinal inorganic ammonia production and absorption - The ammonia (NH_3) in the colon is further converted into ammonium salt (NH4+), which is excreted with urine and feces to alleviate and treat hepatic encephalopathy
Promotes calcium absorption - with the intake of isomerized lactose, the calcium absorption rate increases significantly

SUGAR ALCOHOL & NEW SUGAR SOURCE

■ Erythritol

Erythritol is a new and natural bulk sweetener. It exists in nature at a low level in many fruits and fermented foods such as grapes, melon, mushrooms, and beer.

- Low calorie 0.2Kcal/g
- Tooth friendly
- Low glyceamic response
- High digestive tolerance, No side effects
- Strong cooling effect
- No hygroscopicity
- Masking off-falvor of high-intensity sweeteners

■ Allulose

Allulose is a naturally occurring rare sugar, a new functional sweetener with similar sweetness to sucrose. It has excellent processing characteristics and good stability. It can be used as sugar-free ingredients in beverages, bakery, dairy products, confectionery, pharmaceuticals and other fields.

- Low calorie 0.4Kcal/g
- High tolerance
- Does not affect blood sugar
- Weight control
- Anti-oxidation
- Maillard reaction

■ Blended sweeteners

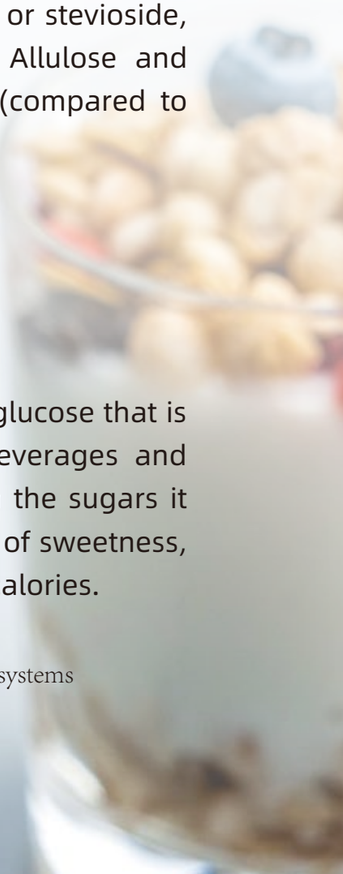
Blended sweeteners are food sweeteners made from a combination of multiple sweeteners. They complement each other to make the taste closer to sucrose with more stable performance and have a synergistic effect on each other. Blended sweeteners can enhance sweetness and flavor, mask bad tastes. Typically two or more natural or artificial sweeteners are blended. Common blends include Erythritol and mogroside or stevioside, which are 1-6 times sweetness (compared to sucrose); Allulose and mogroside or stevioside, which are 1-6 times sweetness (compared to sucrose).

- Synergistic effect, reduce costs
- Functional development and application
- Improve the sweetness stability of sweeteners
- Reduce bad taste, enhance flavor and improve taste

■ Crystalline fructose

Crystalline fructose is a processed sweetener derived from glucose that is almost entirely fructose. It is used as a sweetener in beverages and yogurts, where it substitutes for table sugar. Sweeter than the sugars it replaces, less sugar can be used to produce a desired level of sweetness, resulting in a roughly 5-percent reduction in the amount of calories.

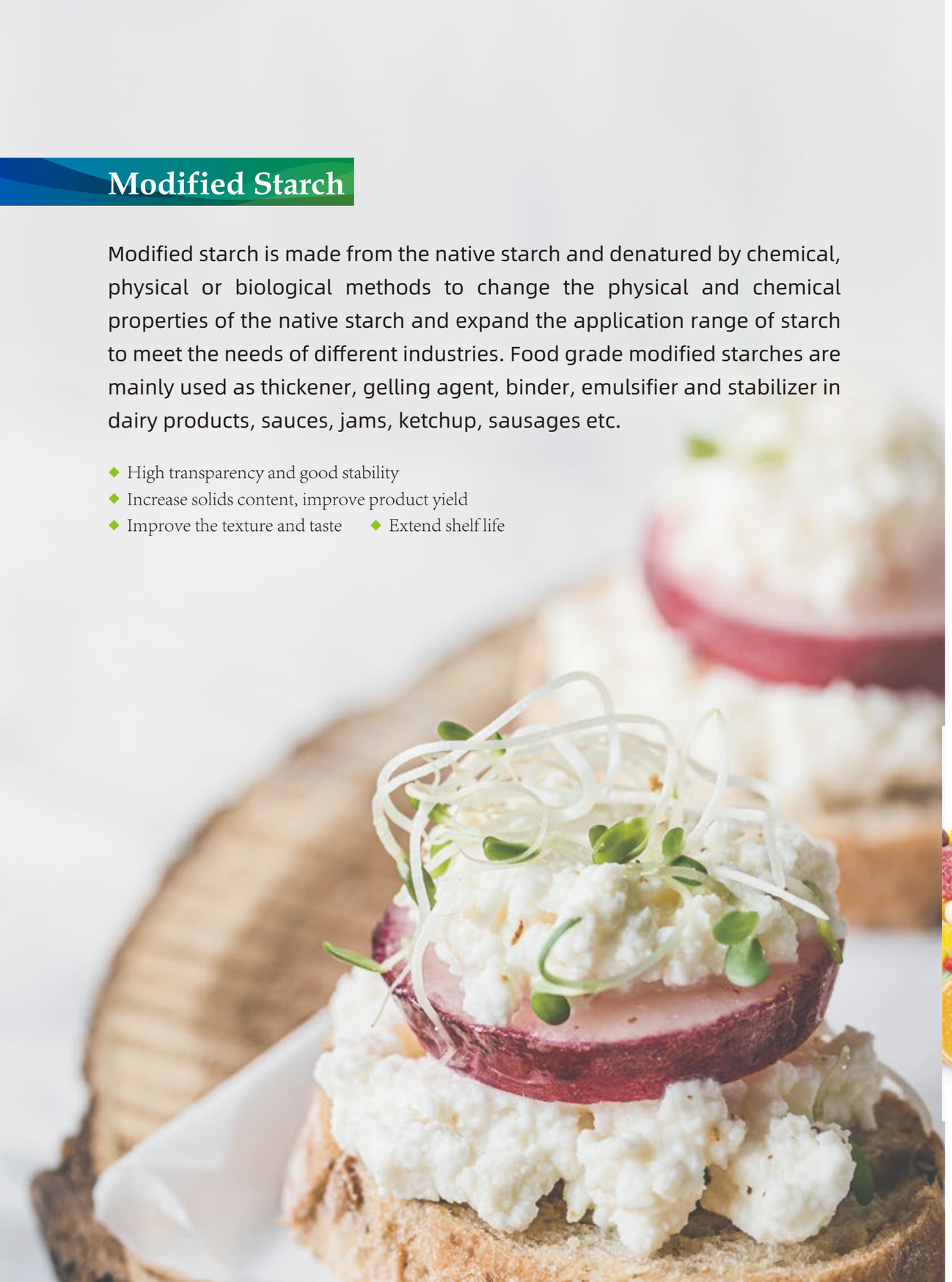
- Higher relative sweetness
- High hygroscopicity
- Greater effect on freezing point depression
- Starch functionality in food systems
- High solubility and difficult to crystallize from an aqueous solution

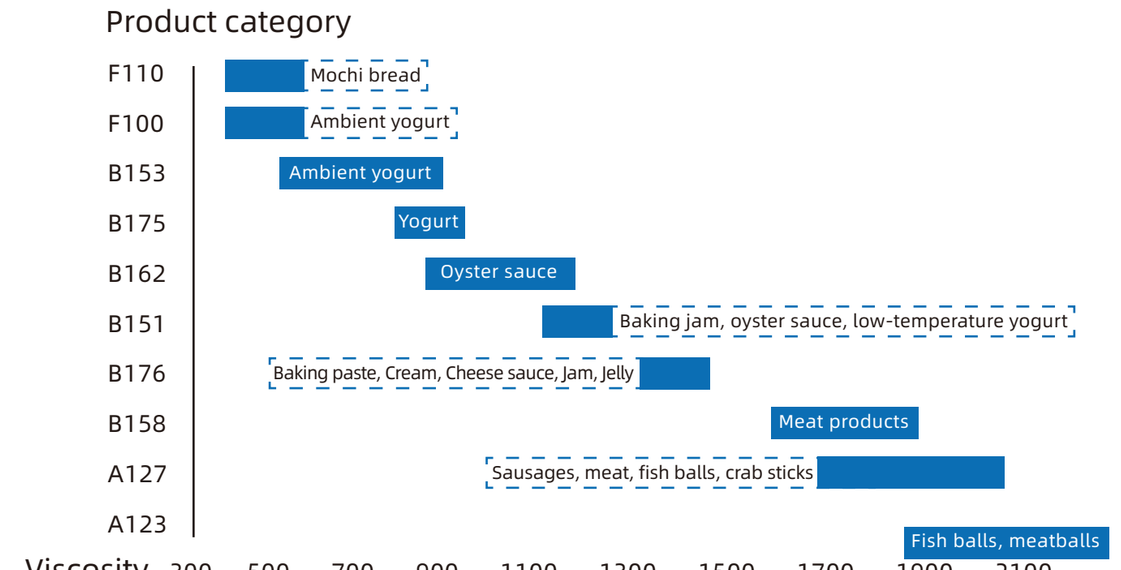


Modified Starch

Modified starch is made from the native starch and denatured by chemical, physical or biological methods to change the physical and chemical properties of the native starch and expand the application range of starch to meet the needs of different industries. Food grade modified starches are mainly used as thickener, gelling agent, binder, emulsifier and stabilizer in dairy products, sauces, jams, ketchup, sausages etc.

- High transparency and good stability
- Increase solids content, improve product yield
- Improve the texture and taste
- Extend shelf life







Recommended products

■ DHA (docosahexaenoic acid)

DHA, or docosahexaenoic acid, is a polyunsaturated fatty acid necessary for the human body, with the molecular formula of C22H32O2, which is a straight-chain fatty acid containing 22 carbon atoms and 6 double bonds. Animal glycerophospholipids contain unequal amounts of DHA, which can be produced from α-linolenic acid during metabolism in vivo, but the amount of production is low, mainly supplemented by food. Docosahexaenoic Acid (DHA, 22:6) is an important member of the Omega-3 (DHA, EPA and ALA, etc.) family, commonly known as brain gold, which supports the health of human beings throughout the life cycle from fetus, infants, young adults to aging. As an important part of the human cerebral cortex and cell membranes in the retina of the eye, DHA is essential for the intelligence and vision development of fetal infants.

- Formula foods for special medical use
- Infant formula
- Health food and health care products
- Functional food
- Pharmaceutical enterprises (API and pharmaceutical excipients)
- Special dietary formula food for infants, pregnant women, and the elderly

■ Isomerized lactose (4-O-β-D-galactose-D-fructofuran)

A disaccharide consisting of galactose and fructose linked by a β-(1→4) glycosidic bond, also known as lactulose. It is produced by isomerizing lactose, which is naturally present in milk.

Industrial production of isomerized lactose is produced by heat treatment of lactose as raw material under alkaline conditions.

- Regulate intestinal flora - isomerized lactose intervention increases the number of probiotics, optimizes the intestinal microenvironment, and relieves constipation
- Reduction of intestinal inorganic ammonia production and absorption – isomerized lactose or rifaximin combined with isomerized lactose can effectively improve HE and reduce mortality
- Promotes calcium absorption – with the intake of isomerized lactose, the calcium absorption rate increases significantly

■ 2’-FL 2'Fucosyllactose (2'-FL)

Human milk oligosaccharides (HMOs) is the third largest solid component after lactose and fat in human milk. They have a variety of structures and functions. Currently, more than 200 HMO structures have been isolated and identified, of which 2'- FL is the most abundant, accounting for about 31%. Studies have shown that infant formula powder added with 2'- FL can significantly reduce the abundance of intestinal pathogens and prevent intestinal infections in babies.

- Immune regulation
- Antiviral activity
- Promotes brain development
- Prevents necrotizing enterocolitis
- Prevents pathogens from sticking to the intestines
- Regulates the intestinal flora in infants as prebiotics

